Cpts 317. 10/21.

Chomsky Normal Form.

last time: CFG & Examples. & Context-free languages

$$S \to 0S1 \mid 01$$

language generated $= \{0^n 1^n : n \geq 1\}$ is a c.f. language

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Show: the following language is c.f.:

$$\mathcal{L} = \{a^n b^m c^m : n \geq 1, m \geq 0\}$$

proof. I will write a c.f. grammar for $\mathcal{L}$:

$$S \to AB$$
$$A \to aA \mid a$$
$$B \to bBc \mid \Lambda$$

Chomsky Normal Form of a CFG:

step 1. Elimination of Λ- production rules:

$$A \longrightarrow \Lambda$$

Example Grammar: I want to eliminate Λ-rules from:

$$S \longrightarrow AASB \mid AS \mid BS \mid ab$$

$$A \longrightarrow AB \mid a \mid \Lambda$$

$$B \longrightarrow BA \mid A \mid b$$

Key: after elimination, the new grammar and the old one generate the same language (minus Λ).

If I only elimate "$A \to \wedge$", then what kind of info that I lose?

Note: $A \rightsquigarrow \wedge$

$B \rightsquigarrow \wedge$

$S \not\rightsquigarrow \wedge$

After I elimate "$A \to \wedge$", then, I lose the info that $A \rightsquigarrow \wedge$.

$B \rightsquigarrow \wedge$.

After elimination:

$$S \longrightarrow AASB \mid ASB \mid SB \mid AAS \mid AS \mid BS \mid ab$$

$$A \longrightarrow AB \mid B \mid a$$

$$B \longrightarrow BA \mid A \mid b$$

One more Example on Λ-elimination:

$$S \longrightarrow ABA \mid AB$$

$$A \longrightarrow aA \mid \Lambda$$

$$B \longrightarrow bB \mid \Lambda$$

Ask yourself: what are the nonterminal symbols that can generate Λ?

$$A \rightsquigarrow \Lambda$$
$$B \rightsquigarrow \Lambda$$
$$S \rightsquigarrow \Lambda$$

Result:

$$S \longrightarrow ABA \mid BA \mid AA \mid AB \mid A \mid B \mid \Lambda$$
$$A \longrightarrow aA \mid a$$
$$B \longrightarrow bB \mid b$$

Step 2. Elimation of Unit production rules:

$$A \longrightarrow B$$

$$S \longrightarrow ABS \mid BA$$
$$A \longrightarrow B \mid a$$
$$B \longrightarrow BA \mid b$$

Note: I will elimate $A \rightarrow B$. what is the info that I lose? we nolonger have:

$$A \longrightarrow BA \mid b$$

Result:

$$S \rightarrow ABS \mid BA$$

$$A \rightarrow a$$

$$B \rightarrow BA \mid b$$

$$A \rightarrow BA \mid b$$

Step 3. Chomsky Normal Form.

Def: A grammar is in CNF if each grammar rule is either in the form of

$$A \longrightarrow BC$$

or

$$A \longrightarrow a$$

We will do an example:

$$S \longrightarrow aS \mid bS \mid AS \mid BS$$

$$A \longrightarrow aAbS \mid AB \mid a$$

$$B \longrightarrow Ba \mid b$$

Conversion:

Fix it.

$E \rightarrow a$

$F \rightarrow b$

$S \rightarrow ES \mid FS \mid AS \mid BS$

$A \rightarrow EAFS \mid AB \mid a$

$B \rightarrow BE \mid b$

$A \rightarrow EH$

$H \rightarrow AI$

$I \rightarrow FS$

One more Example:

$$S \longrightarrow 0S1 \mid 01$$

Conversion:

$$A \longrightarrow 0$$

$$B \longrightarrow 1$$

$$S \longrightarrow \cancel{ASB} \mid AB$$

$$S \longrightarrow AC$$

$$C \longrightarrow SB$$

One more Exmple:

$$S \to AaBB \mid ab$$
$$A \to aBB \mid a$$
$$B \to bAb \mid b$$

Conversion:

$$C \to a$$
$$D \to b$$
$$S \to \cancel{ACBB} \mid CD$$
$$A \to \cancel{CBB} \mid a$$
$$B \to \cancel{DAD} \mid b$$

$$S \to AE$$
$$E \to CF$$
$$F \to BB$$
$$A \to CG$$
$$G \to BB$$
$$B \to DH$$
$$H \to AI$$

Summary:

1. How to write a grammar for a language;
2. Eliminate Λ-rules;
3. Eliminate Unit-rules,
4. Convert a CFG into Chomsky Normal Form

Reg. languages $\longleftrightarrow$ FA

c.f. languages $\longleftrightarrow$ PDA (pushdown automata).

Intuitively, PDA = FA + a pushdown stack.